

Module 02: Agents — Variational Inference and the Recognition Model

Course: Active Inference 101 | **Unit:** Mathematical Frameworks | **Audience:** First-semester undergraduates

Learning Objectives

1. Explain why exact Bayesian inference is usually **intractable** and why we need **approximations**.
2. Define the **recognition model** (approximate posterior) $q(s)$ as the agent's best guess.
3. Introduce **KL divergence** as a measure of the difference between two probability distributions.

Introduction

Module 01 introduced Bayes' theorem as the ideal update rule. There's a problem: for realistic models, exact Bayesian inference is computationally impossible — it requires summing over all possible states of the world. The brain can't do this any more than a computer can. So how does inference work in practice? Through *approximation*.

Key Concepts

1. The Intractability of Exact Inference

Bayes' theorem says: $P(s \mid o) = P(o \mid s) \times P(s) / P(o)$

The denominator $P(o) = \sum_s P(o \mid s) \times P(s)$ requires summing over *every possible state*. For realistic models with many state dimensions, this sum is astronomically large. Even powerful computers can't compute it.

Example: If you have 100 binary hidden states, there are $2^{100} \approx 10^{30}$ possible state combinations — more than the number of atoms in a human body.

2. Variational Inference — The Approximation Strategy

The solution: instead of computing the exact posterior $P(s | o)$, approximate it with a simpler distribution $q(s)$ that is close to the true posterior:

- $q(s)$ is called the **recognition model** or **approximate posterior**
- We choose $q(s)$ from a family of simple distributions (e.g., Gaussians)
- We adjust $q(s)$ to be as close as possible to $P(s | o)$

This is **variational inference**: turning an intractable computation into an optimization problem.

3. KL Divergence — Measuring Distribution Distance

How do we measure "closeness" between $q(s)$ and $P(s | o)$? Using the **Kullback-Leibler (KL) divergence**:

$$D_{KL}[q(s) \parallel P(s | o)] = \sum_s q(s) \times \log[q(s) / P(s | o)]$$

Properties:

- $D_{KL} \geq 0$ always (non-negative)
- $D_{KL} = 0$ if and only if $q(s) = P(s | o)$ (the approximation is exact)
- Not symmetric: $D_{KL}[q \parallel p] \neq D_{KL}[p \parallel q]$ in general

Intuition: KL divergence measures the "surprise" of using distribution q when the true distribution is P . Minimizing KL divergence means making your approximation as unsurprising as possible.

4. The ELBO — Evidence Lower Bound

We can't minimize D_{KL} directly (it requires knowing $P(s | o)$, which is what we're trying to find!). But we can minimize something equivalent: the **negative ELBO** (Evidence Lower Bound):

$$F = -\text{ELBO} = D_{KL}[q(s) \parallel P(s | o)] - \log P(o)$$

Since $\log P(o)$ is constant with respect to q , minimizing F with respect to q is the same as minimizing KL divergence.

The ELBO can be rewritten as: $\text{ELBO} = E_q[\log P(o, s)] - E_q[\log q(s)]$ $\text{ELBO} = \text{Expected accuracy} - \text{Complexity}$

This decomposes into:

- **Accuracy:** How well does q explain the observations?
- **Complexity:** How far is q from the prior $P(s)$?

5. From Variational Inference to Free Energy

The quantity F above is the **Variational Free Energy** — the central quantity in Active Inference. Minimizing F is equivalent to:

- Making $q(s)$ close to the true posterior $P(s | o)$ (good inference)
- Maximizing the ELBO (good model evidence)
- Balancing accuracy and complexity (good modeling)

This is the mathematical formalization of the "free energy minimization" discussed in the Cognitive Science unit.

Summary

Exact Bayesian inference is intractable for realistic models. Variational inference approximates the true posterior with a simpler recognition model $q(s)$, optimized by minimizing KL divergence. The Evidence Lower Bound (ELBO) provides a computable objective. Variational free energy F is the negative ELBO, and minimizing it is the core mathematical operation of Active Inference.

Further Reading

- Blei, D. M., Kucukelbir, A., & McAuliffe, J. D. (2017). Variational inference: A review for statisticians. *JASA*, 112(518), 859-877.

- Buckley, C. L. et al. (2017). The free energy principle for action and perception. *Journal of Mathematical Psychology*, 81, 55-79.
- Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference*. MIT Press. (Chapter 3)